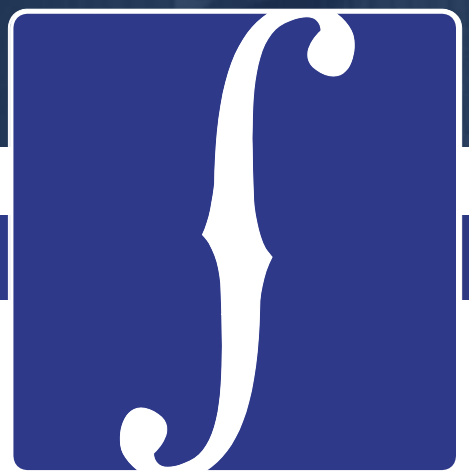




CELLO

Modeling the Forward Evolution of Media Effect:
Model v1.0
July 2016

1

Pool Level Data Set

- Conventional 30-year Fixed-rate Pools (FGLMC and FNCL) Factor data from 1994-01
- We consider this to be a good representation of the Universe of Conventional Agency Mortgages

2

Loan Level Data Set

- Conventional 30-year Fixed-rate Loans (FGLMC) Factor data from 2006-09. (FNCL) Factor data from 2014.
- Loan level data in the early years is only a small fraction of the actual Universe balance (Pool Data)
- By 2016, loans are ~70% of the Universe Balance

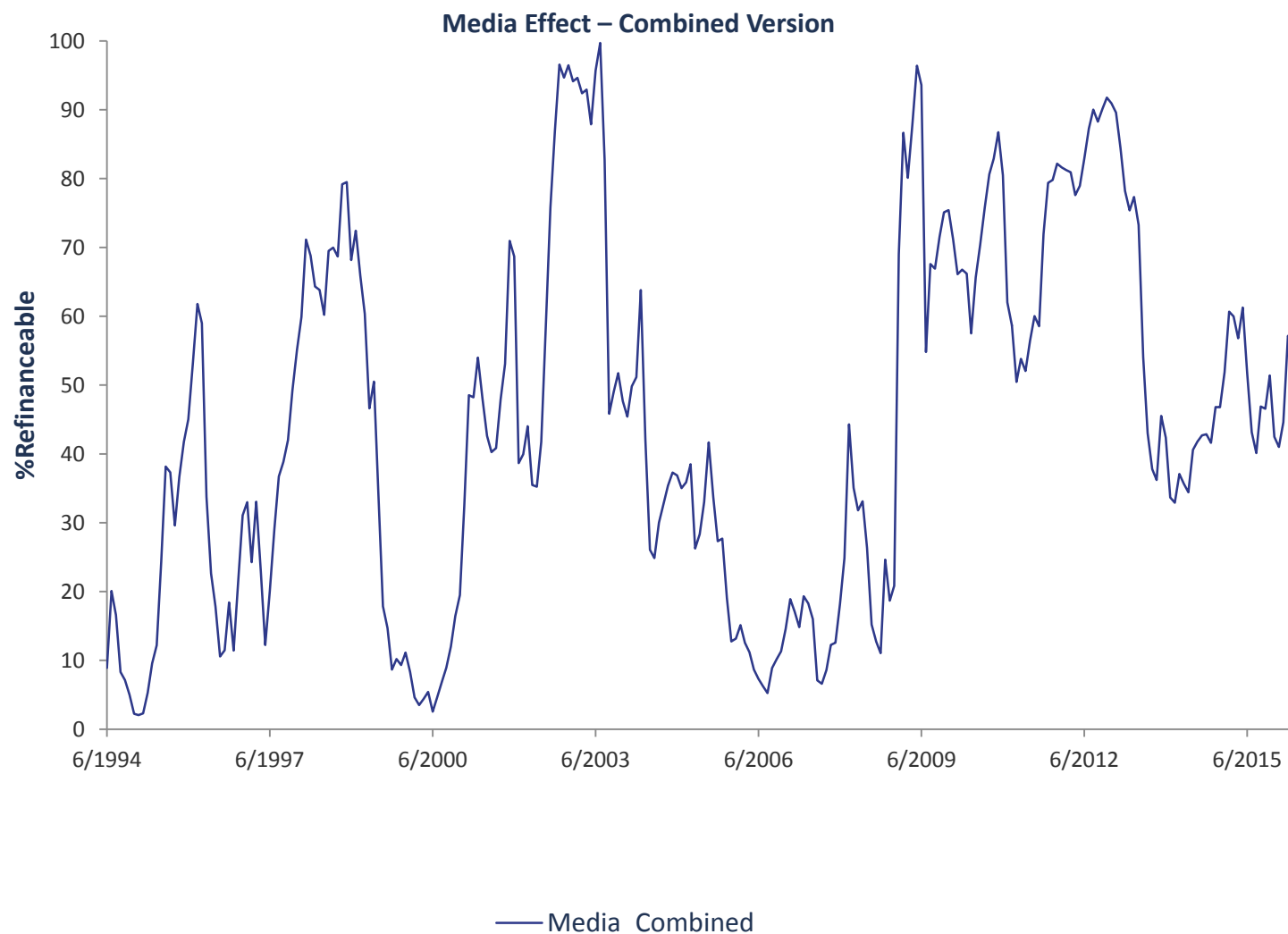
3

Media Effect

- Media Effect = %Refinanceable. Refinanceable = Refi-Incentive > 50 bp
- Refi-Incentive = Note Rate + Original PMI – Current Mtg Rate – Current LLPA – Current PMI
- We combine the Loan Level and Pool Level calculations of the Media Effect to get a combined picture from 1994 to 2016, and more weight is placed on the Loan Level calculation as the Loan Level data set increases

- Media Effect – Combined Loan and Pool History**

To handle the problem that there is no loan level data before 2006, we use the pool level before 2006.



1 Representing the Mortgage Universe

- Create a repline representation of the current universe of Conventional 30-year Fixed-rate Mortgages
- The repline representation consists of WAC buckets and their relative weights
- The repline representation is made available to Scale infrastructure via CDU files for each historical Factor Date

2 Forecasting the Evolution of the Mortgage Universe

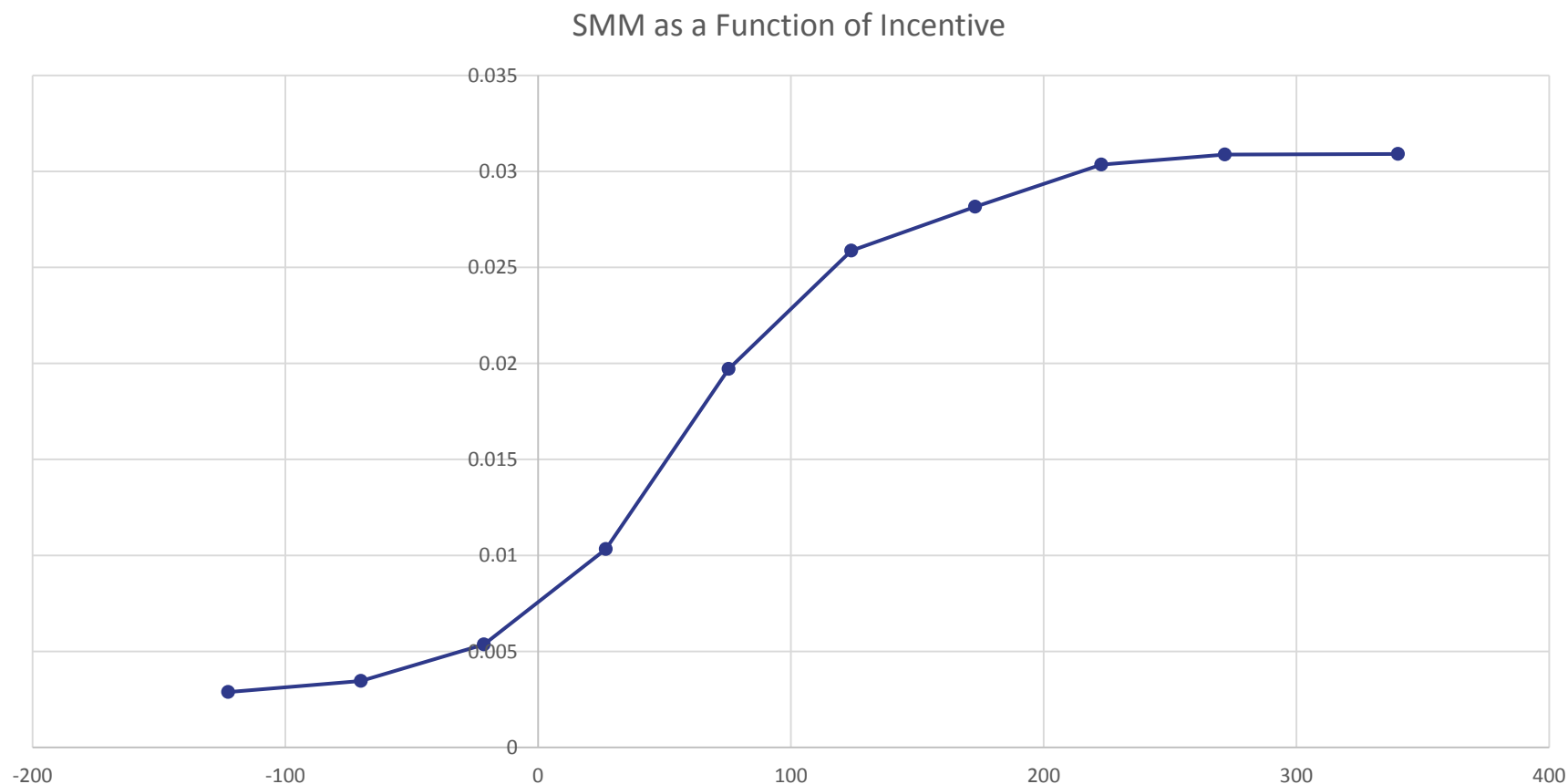
- Calculate the SMM for each WAC bucket of the Universe forward through time
- Option 1: Create a simple S-Curve from empirical data to calculate an SMM based only on Incentive
- Option 2: Call the full prepayment model on each WAC bucket of the Universe¹

3 Updating the Media Effect

- Redistribute balances from each WAC bucket based on these SMMs and the prevailing mortgage rate
- Calculate the Media Effect from the redistributed balances and the prevailing mortgage rate
- Evolve the collateral characteristics of the WAC buckets for the next time step and return to Step 2

¹Note that the Media Effect is also an input into the prepayment model. Given a Media Effect input at $t=0$, the prepayment model produces a 1-step ahead prepayment forecast which is then used to construct the Media Effect for that time period ($t=1$). This level of the Media Effect is then used as an input by the prepayment model to construct the Media Effect for $t=2$ etc.

Population		100			Surviving	Reinvestment	
\$ at t=0	%Dist	WAC Range	Bucket WAC	%CPR	\$ at t=1	\$ at t=1	%Dist
10	10%	6.00-6.25	6.13%	40%	6.0		6%
25	25%	5.75-6.00	5.88%	35%	16.3		16%
25	25%	5.50-5.75	5.63%	30%	17.5	23.8	41%
30	30%	5.25-5.50	5.38%	10%	27.0		27%
10	10%	5.00-5.25	5.13%	5%	9.5		10%
CC		5.12%			5.12%		
Mtg Rate		5.62%			5.62%		
Refi Trigger		6.12%			6.12%		
%Refinanceable		5%			3%		



- A simple S-curve does not accurately model the evolution of the mortgage universe over time.
- **We follow Option 2 in our model for evolving the Media Effect**

1

Including “New” Originations

- The universe of mortgage debt is not static – it is usually growing since an incremental amount of “new” loans are originated at prevailing mortgage rates – these loans correspond to purchases of new homes or an increase in mortgage balances through a cash-out refinance.

2

WAC Bucketing

- Decide how granular the WAC buckets need to be for accuracy
- Test WAC Buckets of different widths – 12.5bps, 25bps, and 50bps

3

Estimating Media Effect Balances

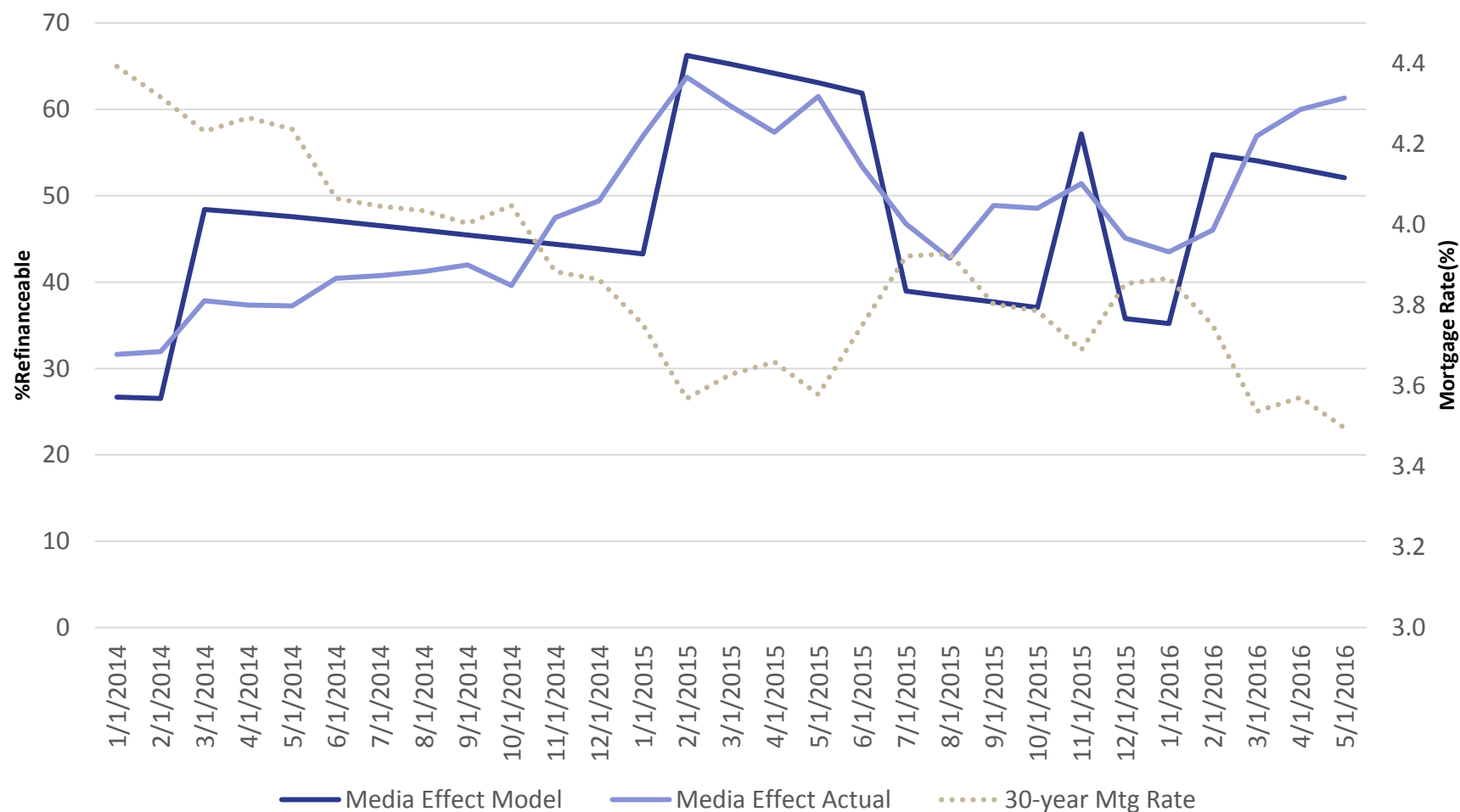
- Simple: If Refi Trigger Rate $\geq$ Avg WAC of the relevant bucket, then full bucket balance is added to Media Balance else only count WAC buckets below
- Average: Linearly accumulate balances based on how the Refi Trigger Rate falls in the relevant WAC bucket

1

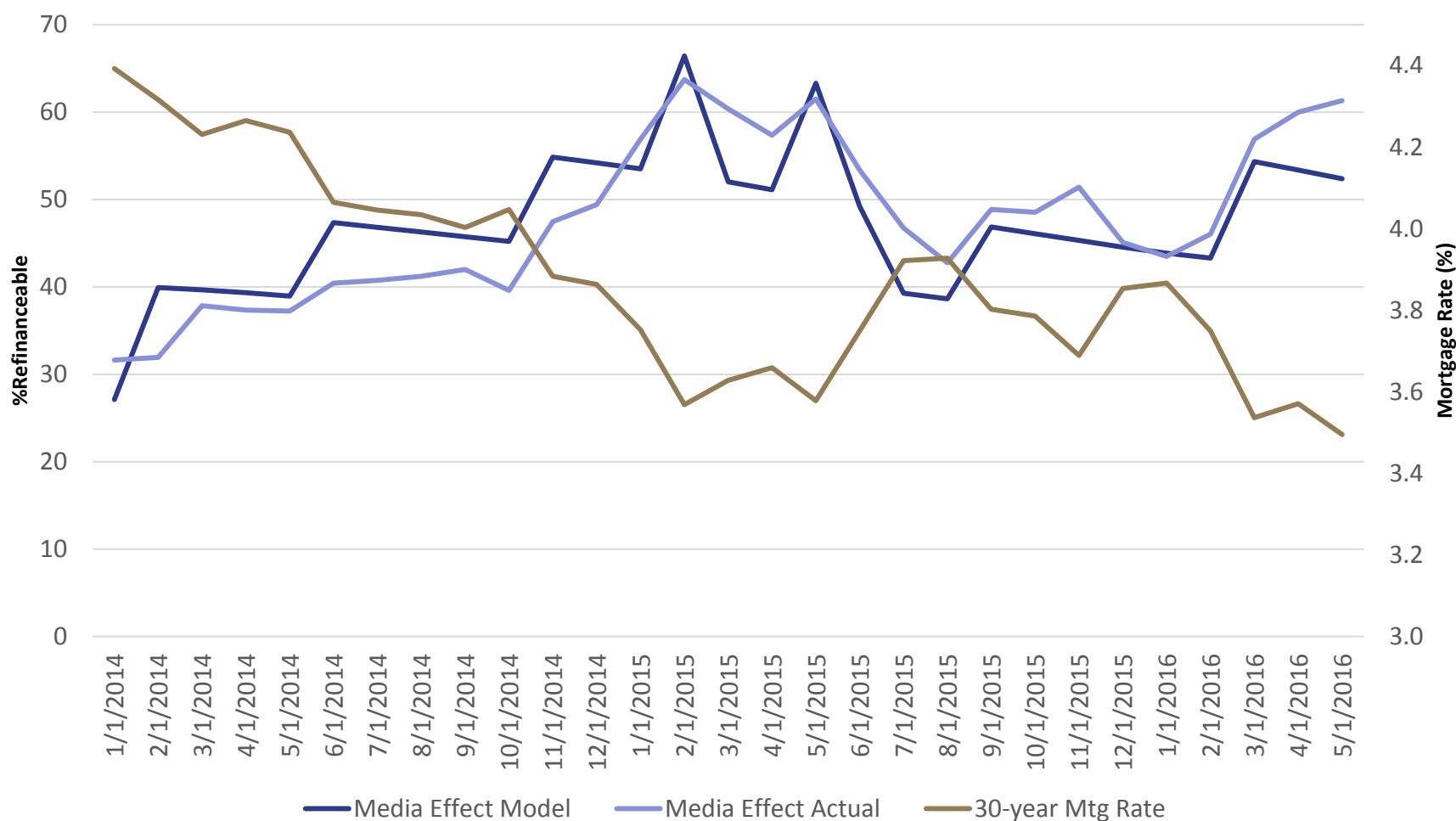
Handling “New” Origination

- The Media Effect Evolution Algorithm in Scale allows for new origination in the appropriate WAC bucket, but does not adjust the WAC bucket attributes: WALA, CLTV, etc.¹
- Test New Origination – 0%, 1% and 2% growth rate (per year)
- Setting New Origination > 0% does not result in better tracking
- **Result = Set New Origination to 0% for this Model**

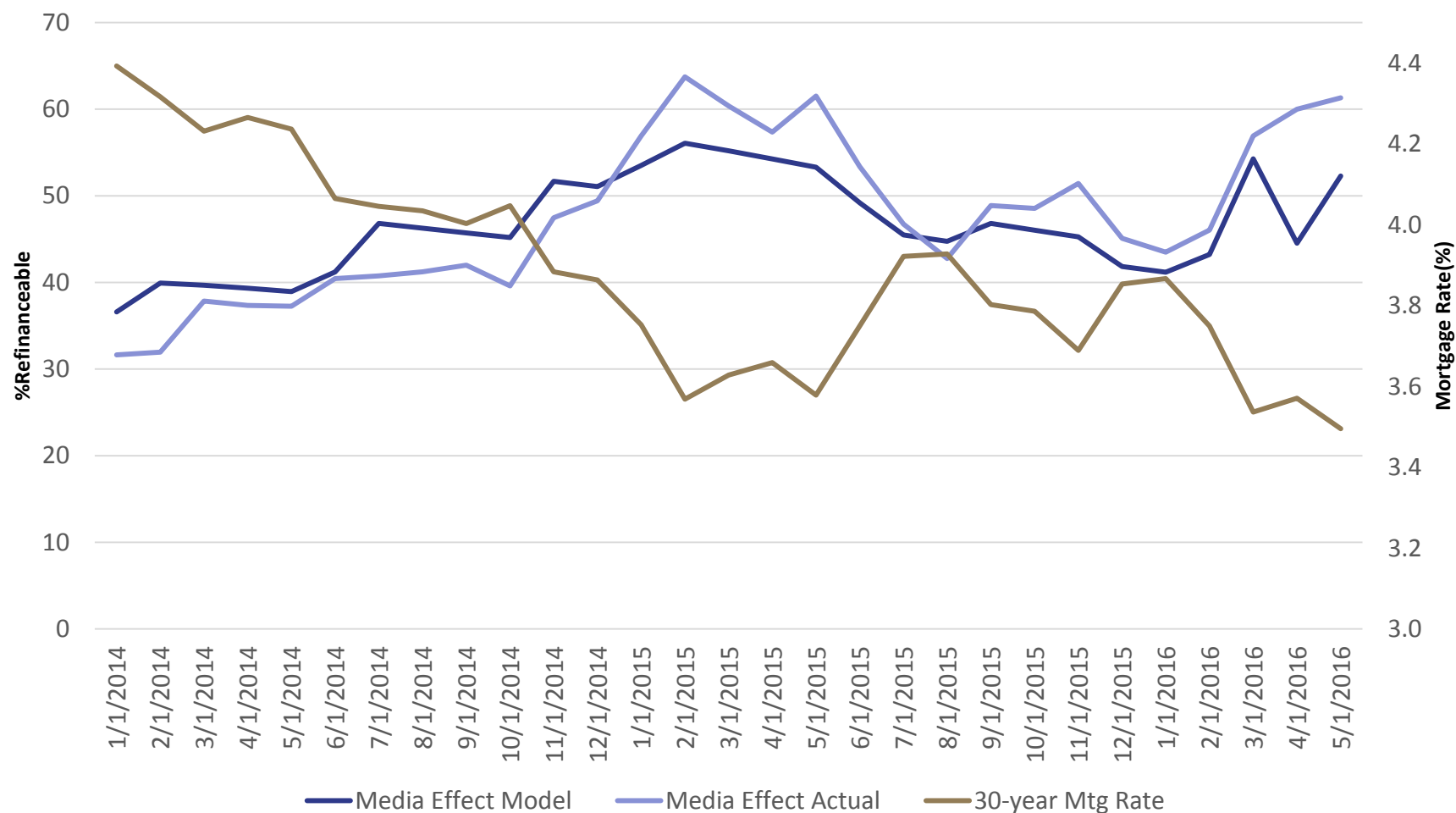
- WAC Bucket Width – 50 bp
- Algorithm – Simple
- RMSE Error = 7.07



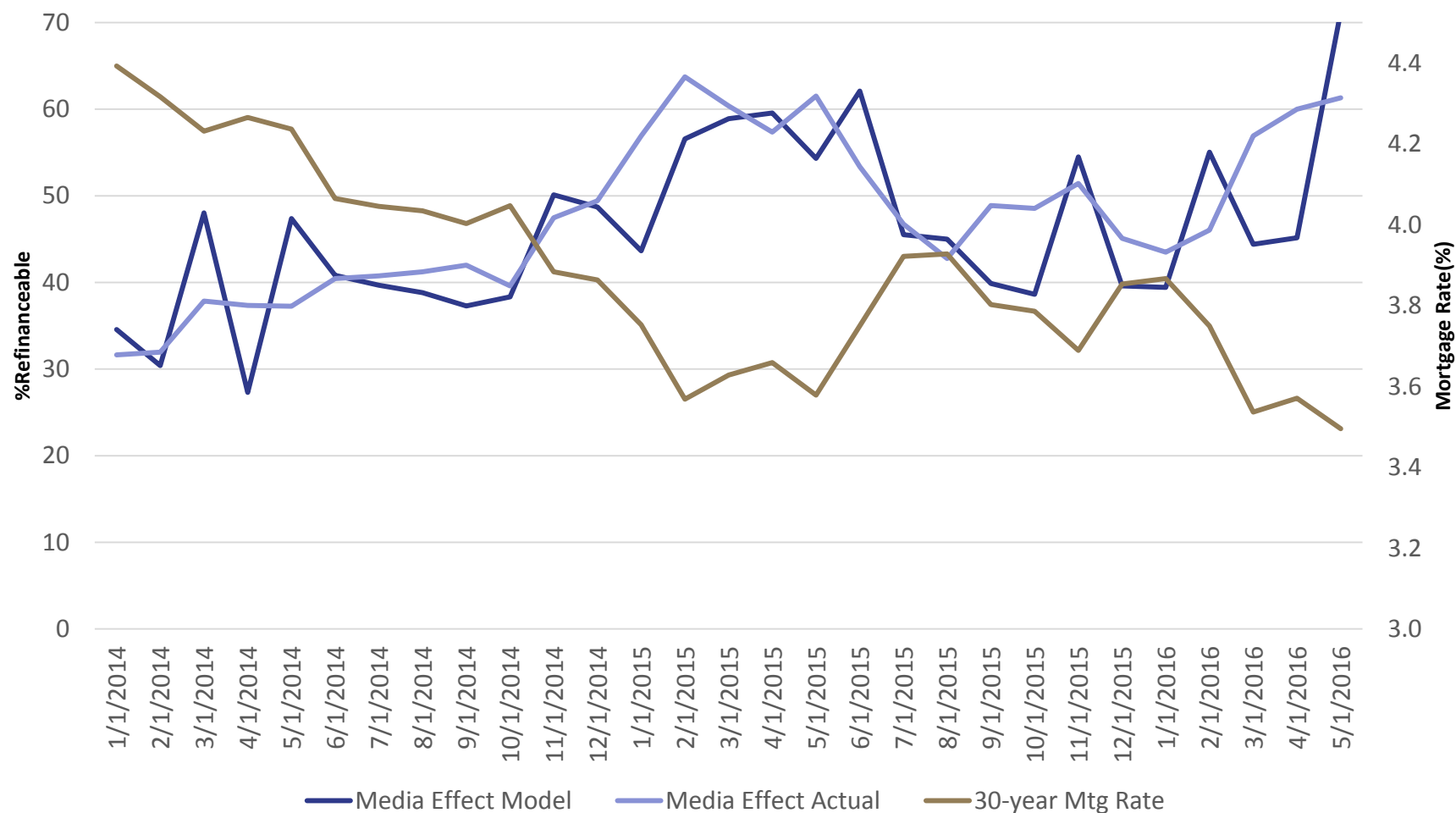
- WAC Bucket Width – 25 bp
- Algorithm – Simple
- RMSE Error = 4.73



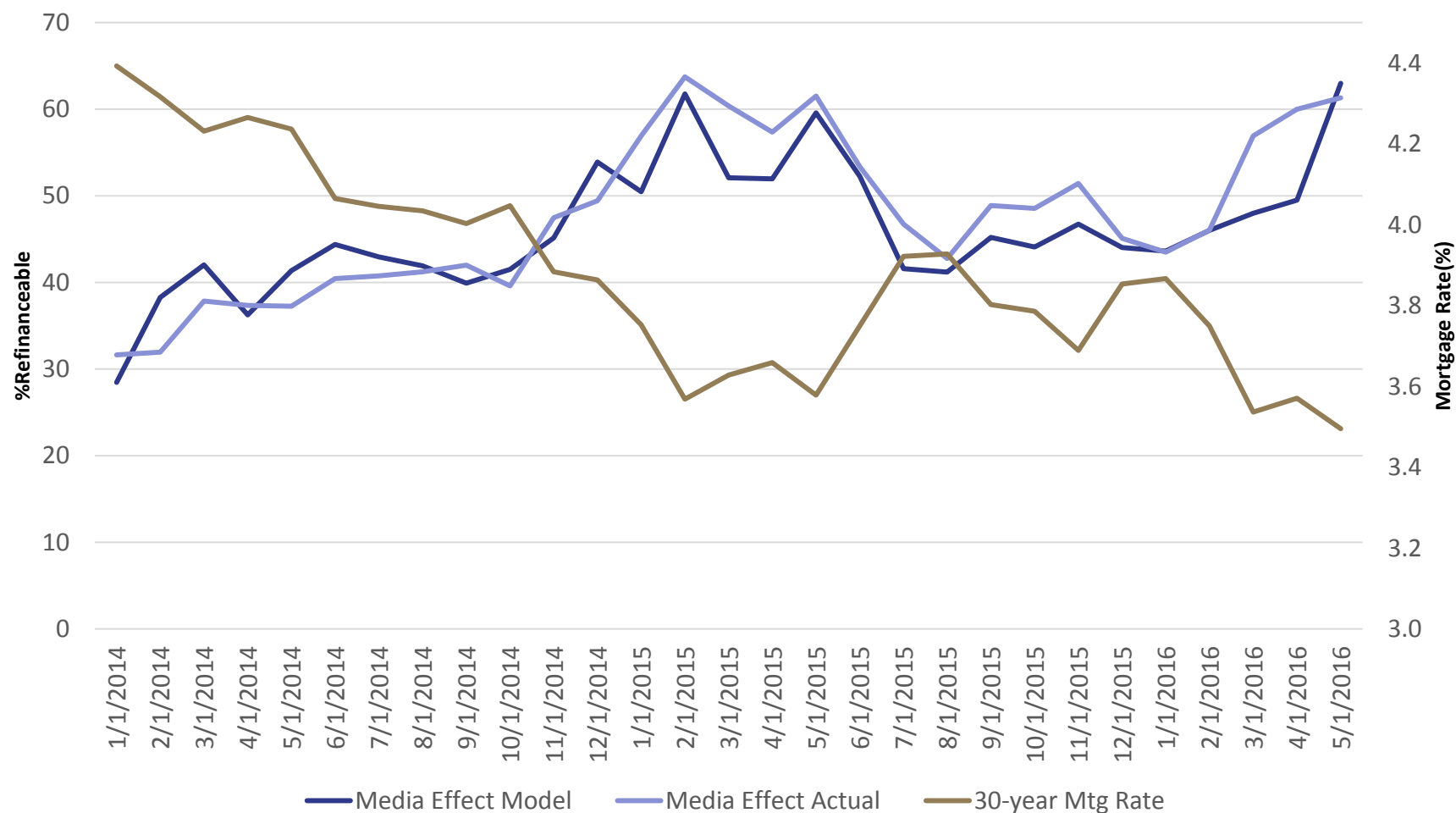
- WAC Bucket Width – 12.5 bp
- Algorithm – Simple
- RMSE Error = 4.49



- WAC Bucket Width – 25 bp
- Algorithm – Average
- RMSE Error = 6.77



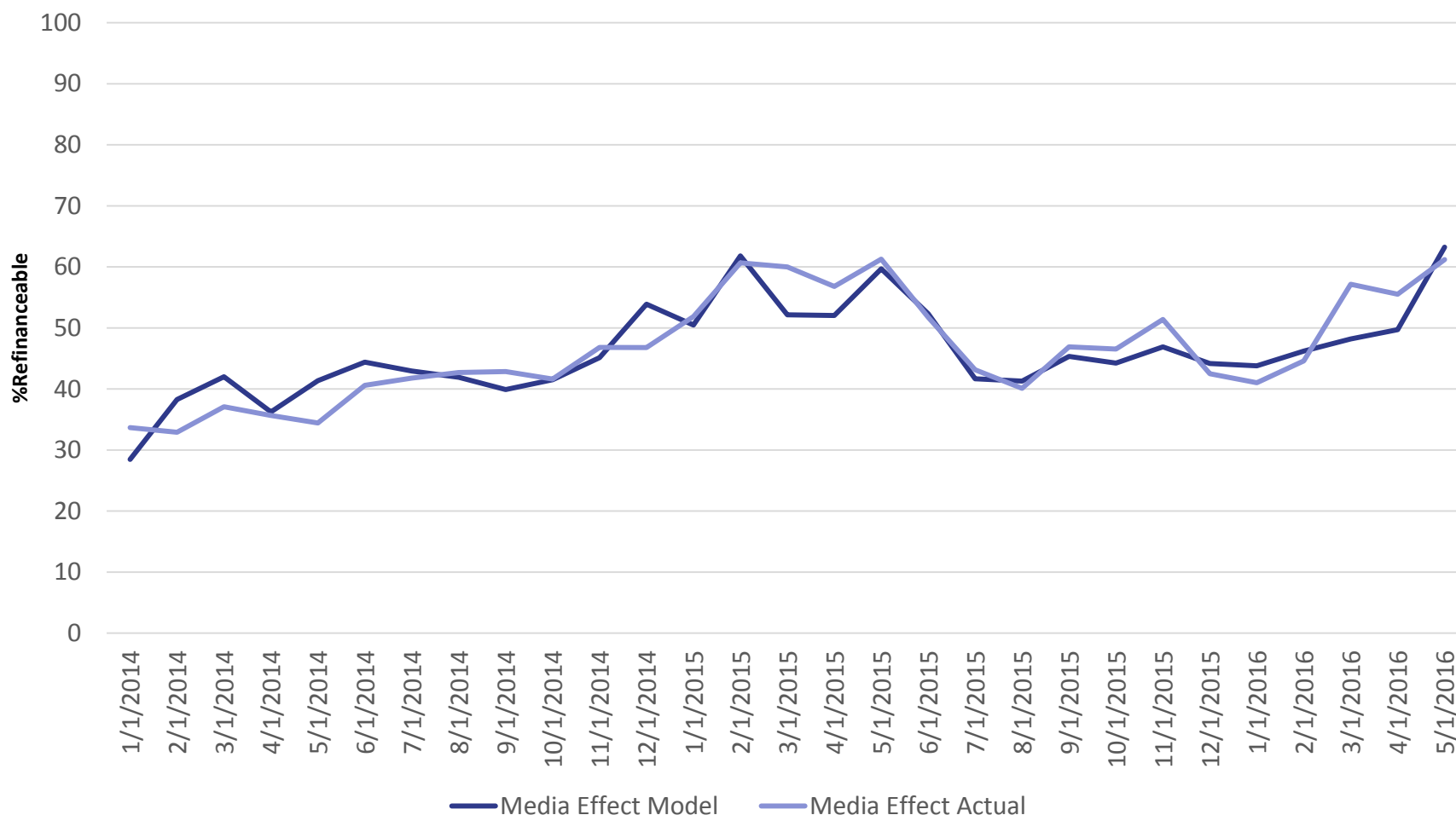
- WAC Bucket Width – 12.5 bp
- Algorithm – Average
- RMSE Error = 3.98



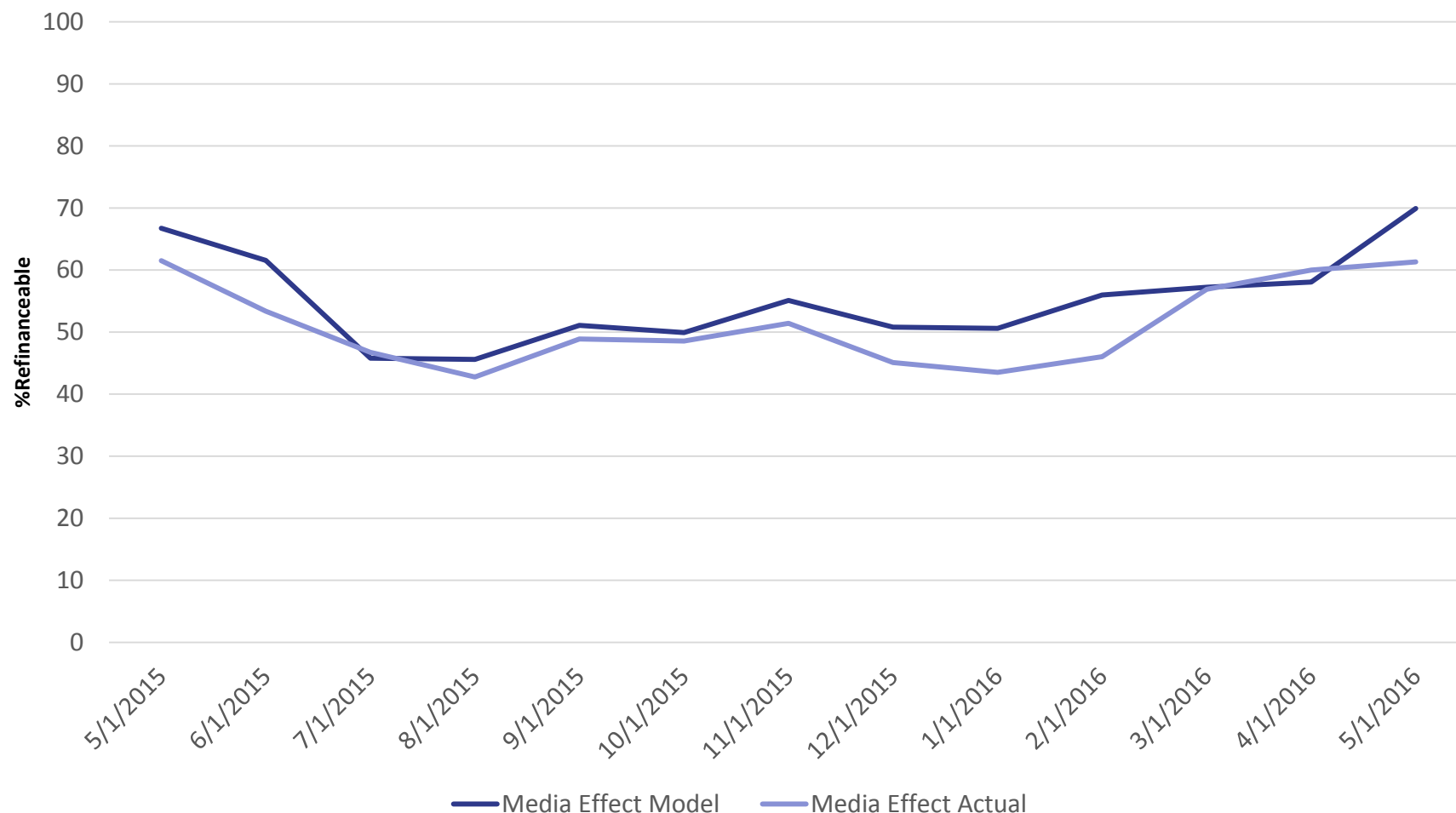
- 2013 to Present
- WAC Bucket – 12.5 bp, Algorithm – Average, 0% New Origination
- RMSE Error = 6.25



- 2014 to Present
- WAC Bucket – 12.5 bp, Algorithm – Average, 0% New Origination
- RMSE Error = 3.98



- 2015-05 to Present
- WAC Bucket – 12.5 bp, Algorithm – Average, 0% New Origination
- RMSE Error = 5.43



1 New Origination

- Set to 0%
- No clear benefit to $> 0\%$

2 WAC Bucketing

- Set to 12.5 BP
- Media Effect Volatility appears consistent with history and tracking is best

3 Media Effect Balance Algorithm

- Set to Average
- Average is better than Simple